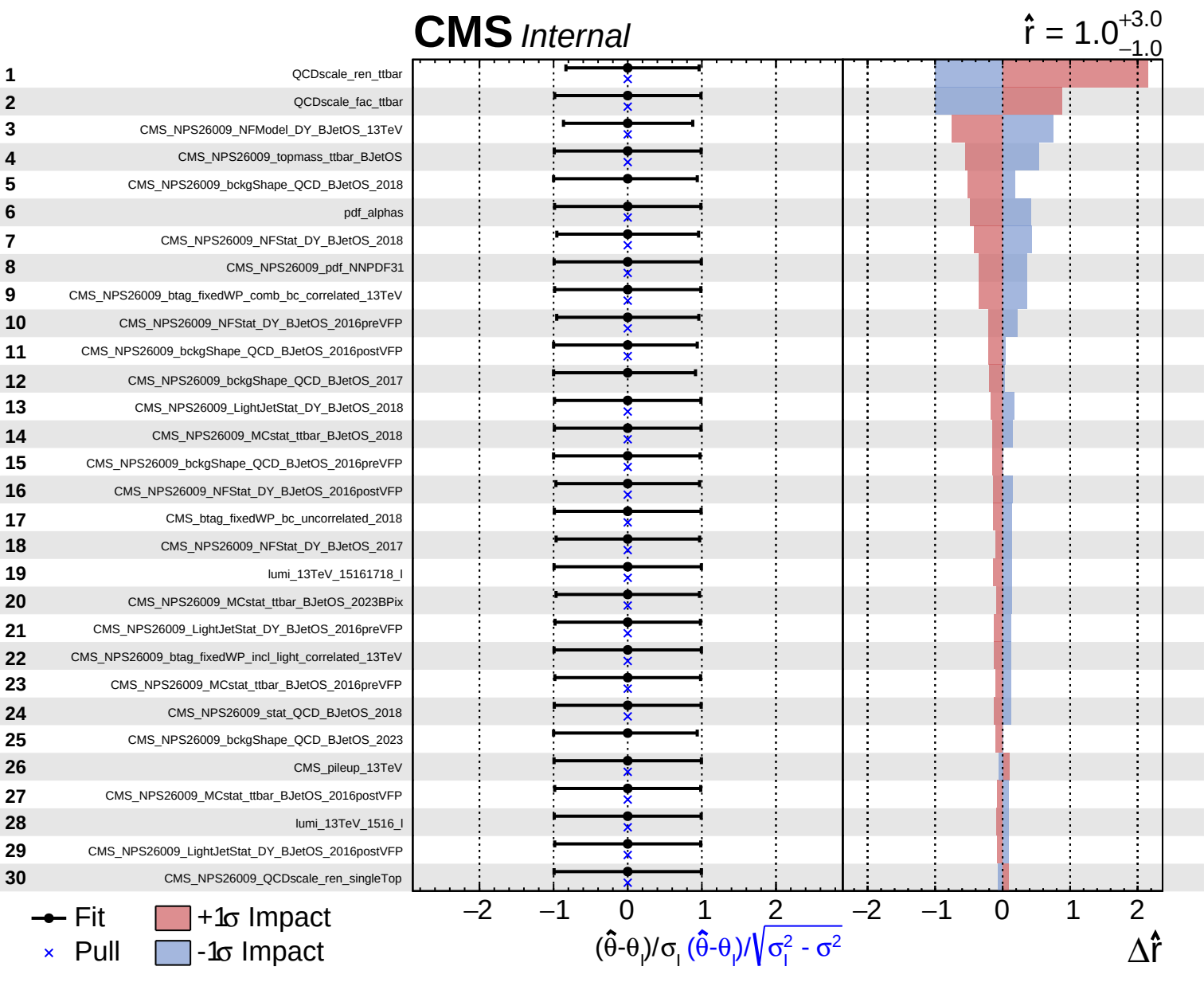
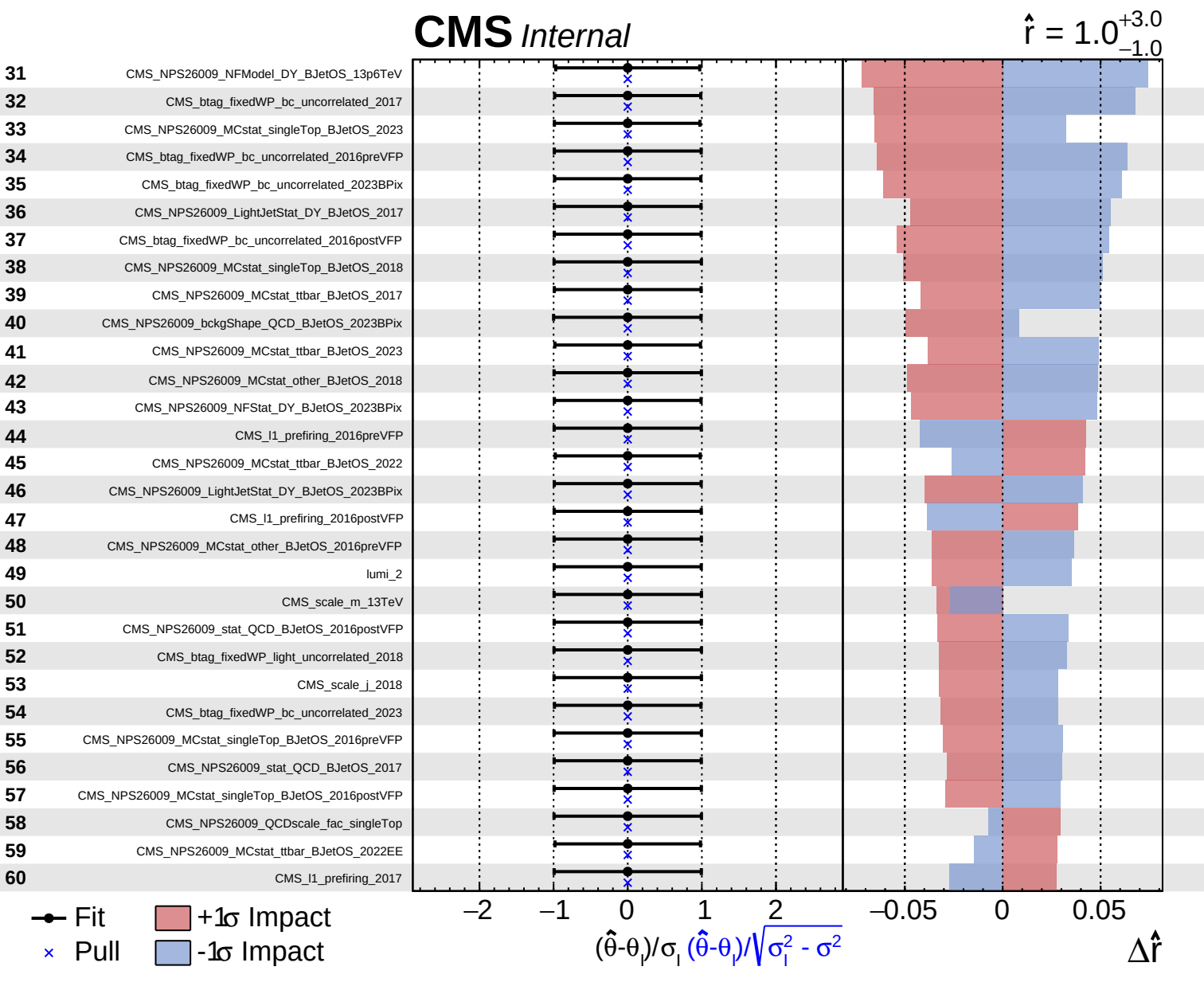
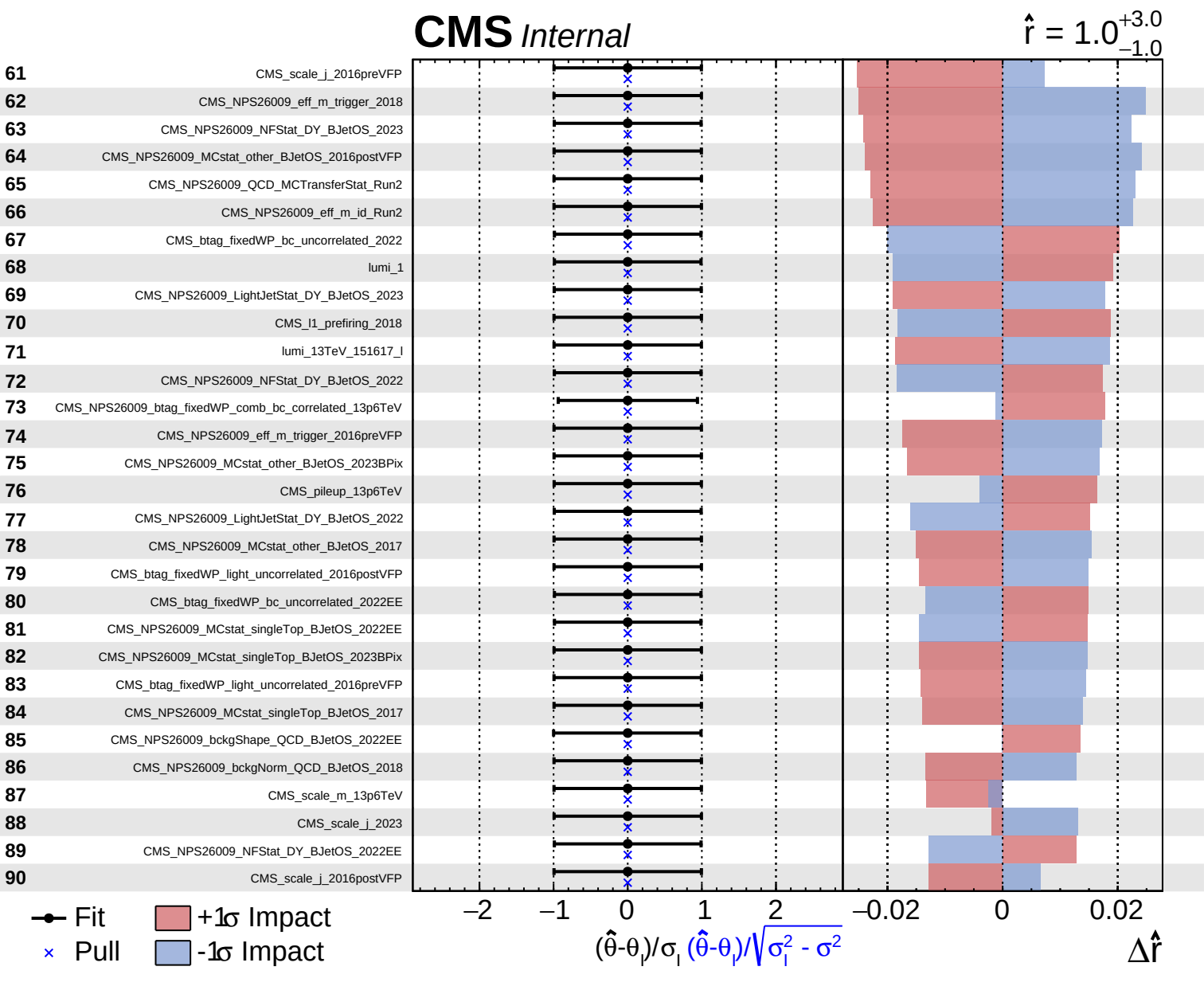
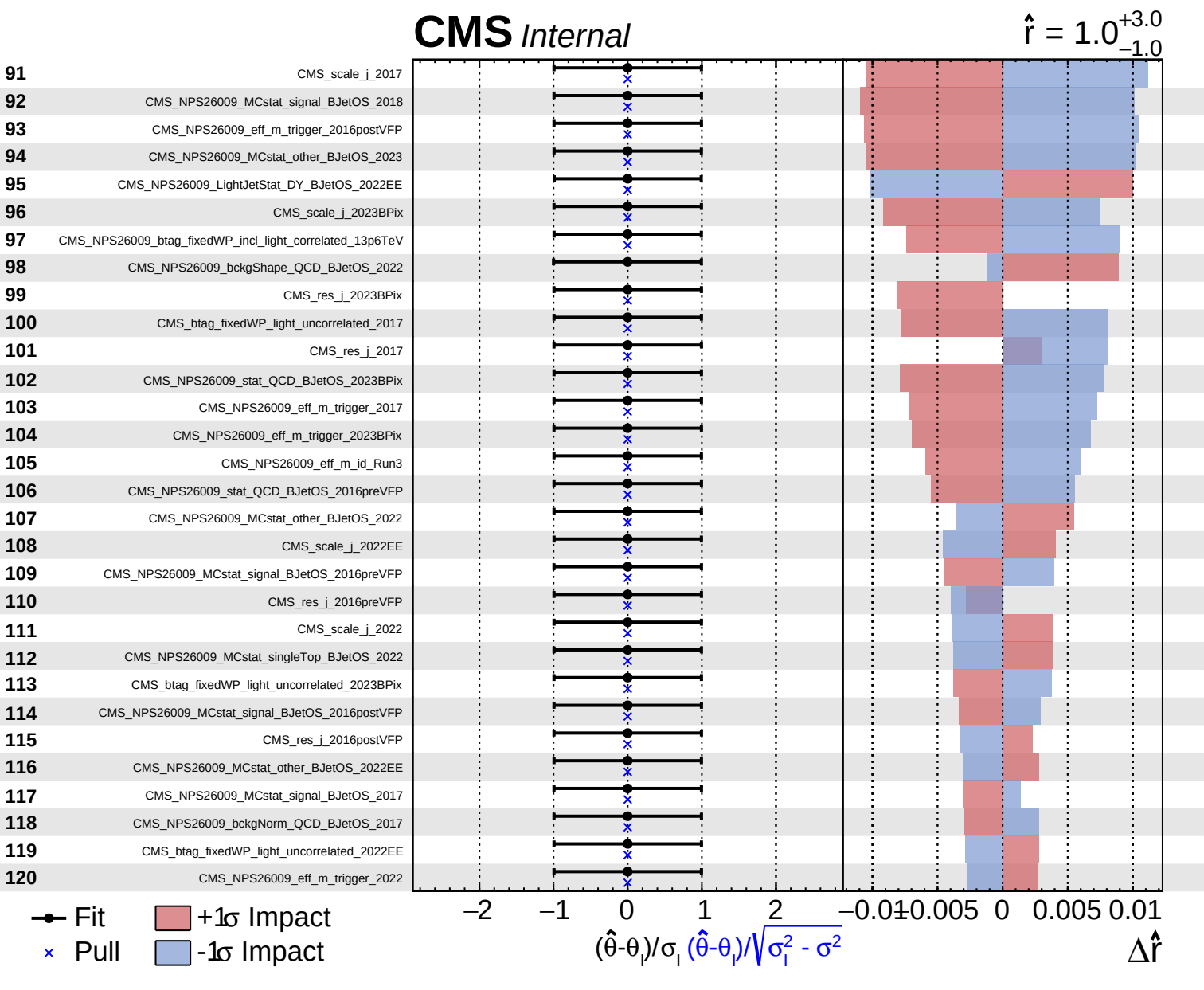
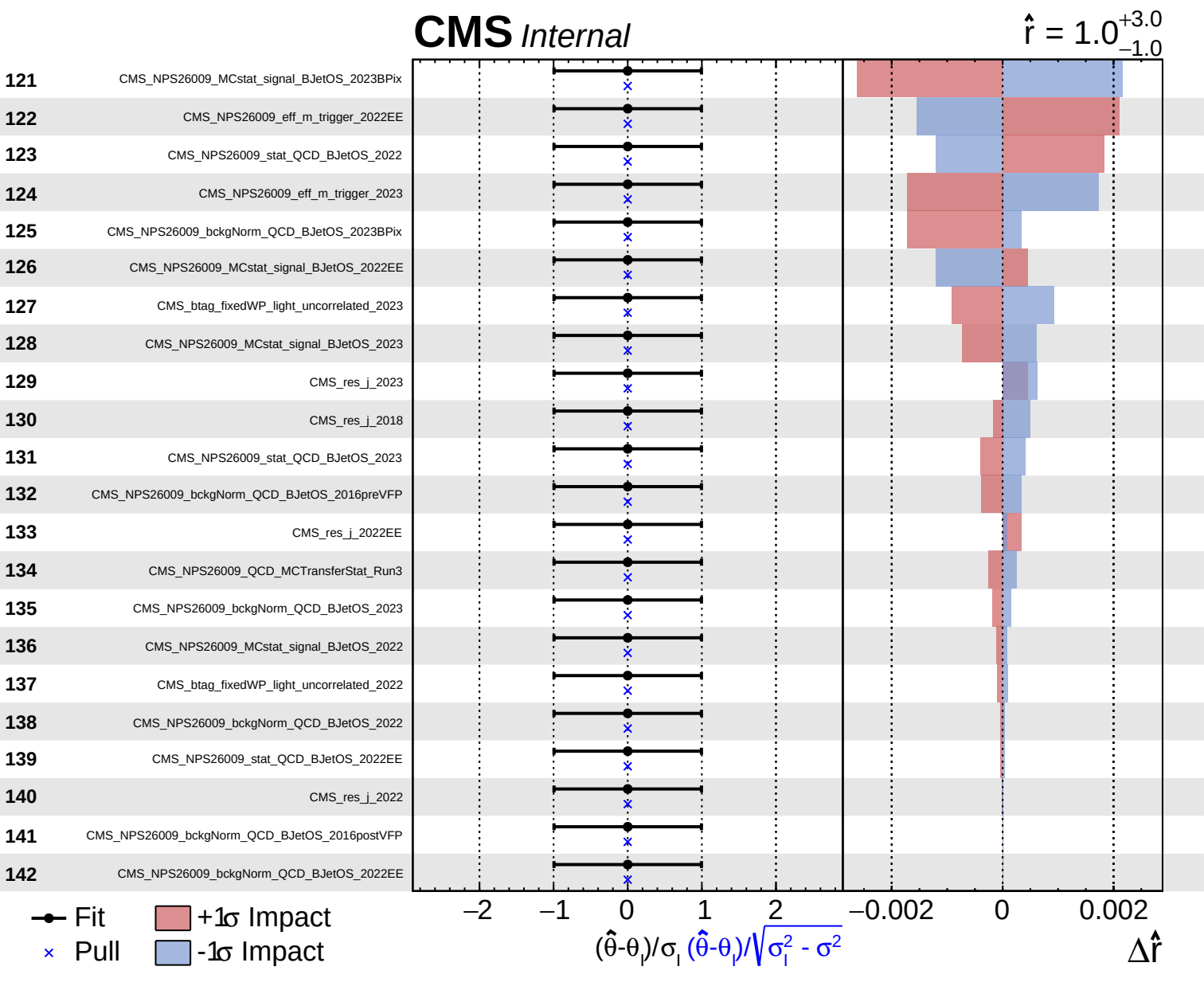

CMS Internal
$\hat{r} = 1.0^{+3.0}_{-1.0}$
1 QCDscale_ren_ttbar
2 QCDscale_fac_ttbar
3 CMS_NPS26009_NFModel_DY_BJetOS_13TeV
4 CMS_NPS26009_topmass_ttbar_BJetOS
5 CMS_NPS26009_bckgShape_QCD_BJetOS_2018
6 pdf_alphas
7 CMS_NPS26009_NFStat_DY_BJetOS_2018
8 CMS_NPS26009_pdf_NNPDF31
9 CMS_NPS26009_btag_fixedWP_comb_bc_correlated_13TeV
10 CMS_NPS26009_NFStat_DY_BJetOS_2016preVFP
11 CMS_NPS26009_bckgShape_QCD_BJetOS_2016postVFP
12 CMS_NPS26009_bckgShape_QCD_BJetOS_2017
13 CMS_NPS26009_LightJetStat_DY_BJetOS_2018
14 CMS_NPS26009_MCstat_ttbar_BJetOS_2018
15 CMS_NPS26009_bckgShape_QCD_BJetOS_2016preVFP
16 CMS_NPS26009_NFStat_DY_BJetOS_2016postVFP
17 CMS_btag_fixedWP_bc_uncorrelated_2018
18 CMS_NPS26009_NFStat_DY_BJetOS_2017
19 lumi_13TeV_15161718_I
20 CMS_NPS26009_MCstat_ttbar_BJetOS_2023BPix
21 CMS_NPS26009_LightJetStat_DY_BJetOS_2016preVFP
22 CMS_NPS26009_btag_fixedWP_incl_light_correlated_13TeV
23 CMS_NPS26009_MCstat_ttbar_BJetOS_2016preVFP
24 CMS_NPS26009_stat_QCD_BJetOS_2018
25 CMS_NPS26009_bckgShape_QCD_BJetOS_2023
26 CMS_pileup_13TeV
27 CMS_NPS26009_MCstat_ttbar_BJetOS_2016postVFP
28 lumi_13TeV_1516_I
29 CMS_NPS26009_LightJetStat_DY_BJetOS_2016postVFP
30 CMS_NPS26009_QCDscale_ren_singleTop
−2 −1 0 1 2
−2 −1 0 1 2
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
r̂ = 1.0 +3.0 −1.0
31 CMS_NPS26009_NFModel_DY_BJetOS_13p6TeV
32 CMS_btag_fixedWP_bc_uncorrelated_2017
33 CMS_NPS26009_MCstat_singleTop_BJetOS_2023
34 CMS_btag_fixedWP_bc_uncorrelated_2016preVFP
35 CMS_btag_fixedWP_bc_uncorrelated_2023BPix
36 CMS_NPS26009_LightJetStat_DY_BJetOS_2017
37 CMS_btag_fixedWP_bc_uncorrelated_2016postVFP
38 CMS_NPS26009_MCstat_singleTop_BJetOS_2018
39 CMS_NPS26009_MCstat_ttbar_BJetOS_2017
40 CMS_NPS26009_bckgShape_QCD_BJetOS_2023BPix
41 CMS_NPS26009_MCstat_ttbar_BJetOS_2023
42 CMS_NPS26009_MCstat_other_BJetOS_2018
43 CMS_NPS26009_NFStat_DY_BJetOS_2023BPix
44 CMS_l1_prefiring_2016preVFP
45 CMS_NPS26009_MCstat_ttbar_BJetOS_2022
46 CMS_NPS26009_LightJetStat_DY_BJetOS_2023BPix
47 CMS_l1_prefiring_2016postVFP
48 CMS_NPS26009_MCstat_other_BJetOS_2016preVFP
49 lumi_2
50 CMS_scale_m_13TeV
51 CMS_NPS26009_stat_QCD_BJetOS_2016postVFP
52 CMS_btag_fixedWP_light_uncorrelated_2018
53 CMS_scale_j_2018
54 CMS_btag_fixedWP_bc_uncorrelated_2023
55 CMS_NPS26009_MCstat_singleTop_BJetOS_2016preVFP
56 CMS_NPS26009_stat_QCD_BJetOS_2017
57 CMS_NPS26009_MCstat_singleTop_BJetOS_2016postVFP
58 CMS_NPS26009_QCDscale_fac_singleTop
59 CMS_NPS26009_MCstat_ttbar_BJetOS_2022EE
60 CMS_l1_prefiring_2017
Fit
Pull
+1σ Impact
-1σ Impact
−2 −1 0 1 2
(θ̂-θ_I)/σ_I
−0.05 0 0.05
(θ̂-θ_I)/√(σ_I² - σ²)
Δr̂


CMS Internal
r̂ = 1.0 +3.0 −1.0
61 CMS_scale_j_2016preVFP
62 CMS_NPS26009_eff_m_trigger_2018
63 CMS_NPS26009_NFStat_DY_BJetOS_2023
64 CMS_NPS26009_MCstat_other_BJetOS_2016postVFP
65 CMS_NPS26009_QCD_MCTransferStat_Run2
66 CMS_NPS26009_eff_m_id_Run2
67 CMS_btag_fixedWP_bc_uncorrelated_2022
68 lumi_1
69 CMS_NPS26009_LightJetStat_DY_BJetOS_2023
70 CMS_l1_prefiring_2018
71 lumi_13TeV_151617_l
72 CMS_NPS26009_NFStat_DY_BJetOS_2022
73 CMS_NPS26009_btag_fixedWP_comb_bc_correlated_13p6TeV
74 CMS_NPS26009_eff_m_trigger_2016preVFP
75 CMS_NPS26009_MCstat_other_BJetOS_2023BPix
76 CMS_pileup_13p6TeV
77 CMS_NPS26009_LightJetStat_DY_BJetOS_2022
78 CMS_NPS26009_MCstat_other_BJetOS_2017
79 CMS_btag_fixedWP_light_uncorrelated_2016postVFP
80 CMS_btag_fixedWP_bc_uncorrelated_2022EE
81 CMS_NPS26009_MCstat_singleTop_BJetOS_2022EE
82 CMS_NPS26009_MCstat_singleTop_BJetOS_2023BPix
83 CMS_btag_fixedWP_light_uncorrelated_2016preVFP
84 CMS_NPS26009_MCstat_singleTop_BJetOS_2017
85 CMS_NPS26009_bckgShape_QCD_BJetOS_2022EE
86 CMS_NPS26009_bckgNorm_QCD_BJetOS_2018
87 CMS_scale_m_13p6TeV
88 CMS_scale_j_2023
89 CMS_NPS26009_NFStat_DY_BJetOS_2022EE
90 CMS_scale_j_2016postVFP
−2
−1
0
1
2
−0.02
0
0.02
Fit
Pull
+1σ Impact
-1σ Impact
(θ̂-θ_I)/σ_I
(θ̂-θ_I)/√(σ_I² - σ²)
Δr̂


CMS Internal
$\hat{r} = 1.0^{+3.0}_{-1.0}$
91 CMS_scale_j_2017
92 CMS_NPS26009_MCstat_signal_BJetOS_2018
93 CMS_NPS26009_eff_m_trigger_2016postVFP
94 CMS_NPS26009_MCstat_other_BJetOS_2023
95 CMS_NPS26009_LightJetStat_DY_BJetOS_2022EE
96 CMS_scale_j_2023BPix
97 CMS_NPS26009_btag_fixedWP_incl_light_correlated_13p6TeV
98 CMS_NPS26009_bckgShape_QCD_BJetOS_2022
99 CMS_res_j_2023BPix
100 CMS_btag_fixedWP_light_uncorrelated_2017
101 CMS_res_j_2017
102 CMS_NPS26009_stat_QCD_BJetOS_2023BPix
103 CMS_NPS26009_eff_m_trigger_2017
104 CMS_NPS26009_eff_m_trigger_2023BPix
105 CMS_NPS26009_eff_m_id_Run3
106 CMS_NPS26009_stat_QCD_BJetOS_2016preVFP
107 CMS_NPS26009_MCstat_other_BJetOS_2022
108 CMS_scale_j_2022EE
109 CMS_NPS26009_MCstat_signal_BJetOS_2016preVFP
110 CMS_res_j_2016preVFP
111 CMS_scale_j_2022
112 CMS_NPS26009_MCstat_singleTop_BJetOS_2022
113 CMS_btag_fixedWP_light_uncorrelated_2023BPix
114 CMS_NPS26009_MCstat_signal_BJetOS_2016postVFP
115 CMS_res_j_2016postVFP
116 CMS_NPS26009_MCstat_other_BJetOS_2022EE
117 CMS_NPS26009_MCstat_signal_BJetOS_2017
118 CMS_NPS26009_bckgNorm_QCD_BJetOS_2017
119 CMS_btag_fixedWP_light_uncorrelated_2022EE
120 CMS_NPS26009_eff_m_trigger_2022
Fit
Pull
+1σ Impact
-1σ Impact
−2 −1 0 1 2
$(\hat{\theta}-\theta_I)/\sigma_I$
−0.01 −0.005 0 0.005 0.01
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
r̂ = 1.0 +3.0 −1.0
121 CMS_NPS26009_MCstat_signal_BJetOS_2023BPix
122 CMS_NPS26009_eff_m_trigger_2022EE
123 CMS_NPS26009_stat_QCD_BJetOS_2022
124 CMS_NPS26009_eff_m_trigger_2023
125 CMS_NPS26009_bckgNorm_QCD_BJetOS_2023BPix
126 CMS_NPS26009_MCstat_signal_BJetOS_2022EE
127 CMS_btag_fixedWP_light_uncorrelated_2023
128 CMS_NPS26009_MCstat_signal_BJetOS_2023
129 CMS_res_j_2023
130 CMS_res_j_2018
131 CMS_NPS26009_stat_QCD_BJetOS_2023
132 CMS_NPS26009_bckgNorm_QCD_BJetOS_2016preVFP
133 CMS_res_j_2022EE
134 CMS_NPS26009_QCD_MCTransferStat_Run3
135 CMS_NPS26009_bckgNorm_QCD_BJetOS_2023
136 CMS_NPS26009_MCstat_signal_BJetOS_2022
137 CMS_btag_fixedWP_light_uncorrelated_2022
138 CMS_NPS26009_bckgNorm_QCD_BJetOS_2022
139 CMS_NPS26009_stat_QCD_BJetOS_2022EE
140 CMS_res_j_2022
141 CMS_NPS26009_bckgNorm_QCD_BJetOS_2016postVFP
142 CMS_NPS26009_bckgNorm_QCD_BJetOS_2022EE
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
(θ̂-θ_I)/σ_I
−0.002
0
0.002
(θ̂-θ_I)/√(σ_I² - σ²)
Δr̂